

Graphs Of Inequalities

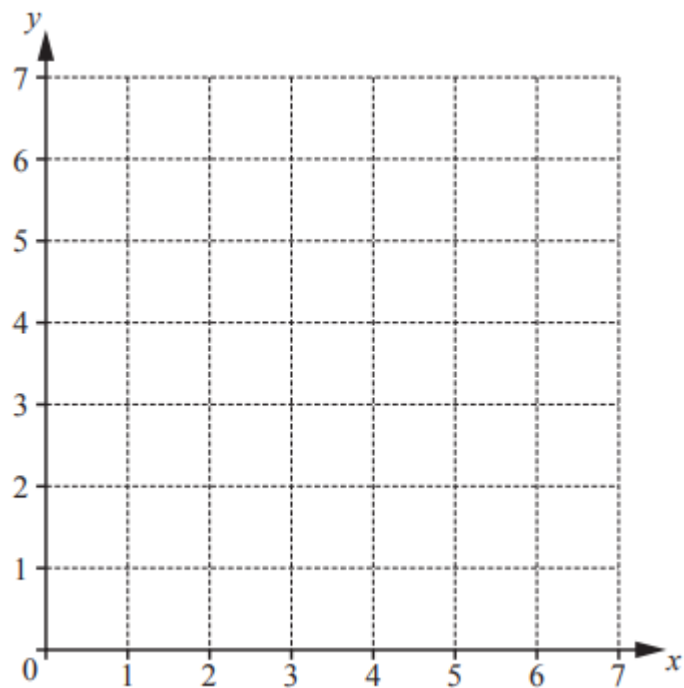
Worksheet 1(Non-Calculator)

1. W15/qp21/q5(c)

- i. Shade and label the region R defined by these four inequalities.

[3]

$$x \geq 1 \quad y \leq 4 \quad x + y \leq 6 \quad y \geq x$$



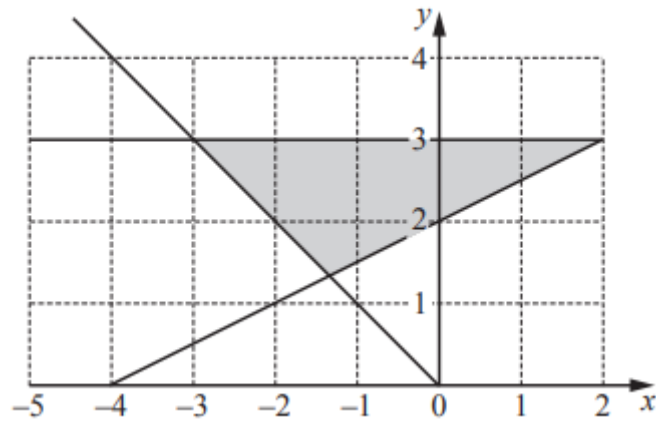
- ii. The point M is the intersection of $x = 1$ and $y = 4$.

The point N is the intersection of $x + y = 6$ and $y = x$.

Find the gradient of MN.

[2]

2. S16/qp11/q9

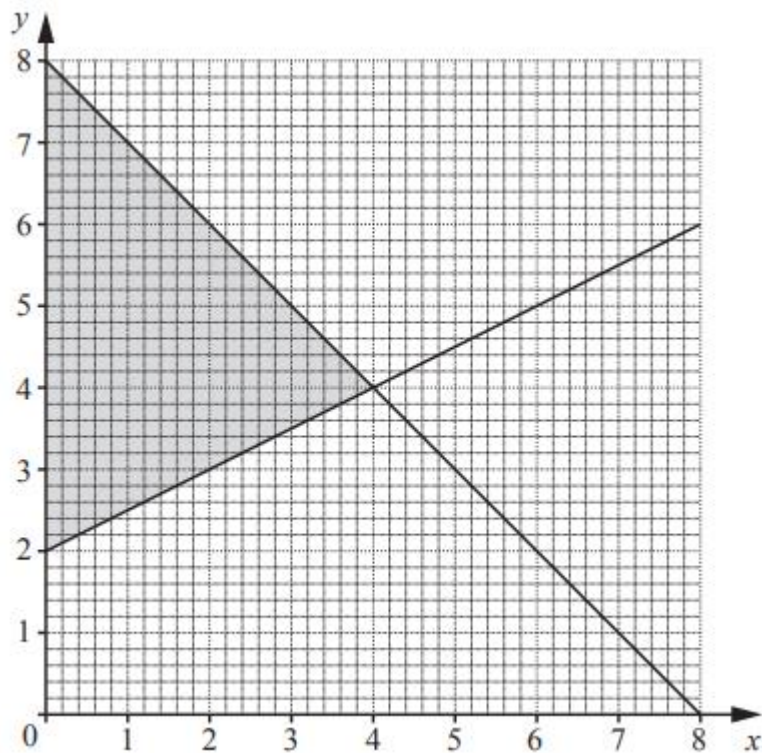


The shaded region in the diagram is defined by three inequalities.

One of these is $3y \geq \frac{1}{2}x + 2$. Write down the other two inequalities.

[2]

3. S16/qp12/q15



The diagram shows the lines $x + y = 8$ and $2y = x + 4$.

a. The shaded region on the diagram is defined by three inequalities.

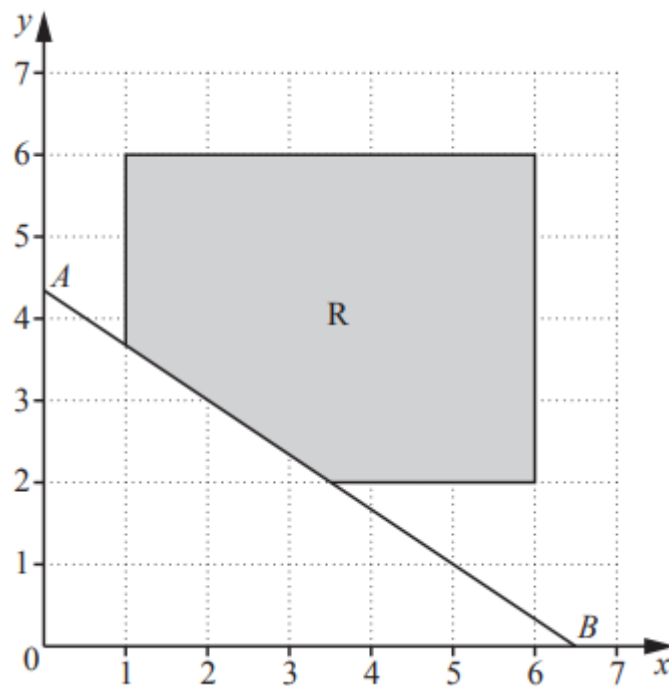
Write down these three inequalities.

[2]

b. Another region, R, is defined by the inequalities $x + y \leq 8$, $2y \leq x + 4$ and $y \geq a$, where a is an integer. This region contains 5 points with integer coordinates. Write down the value of a.

[1]

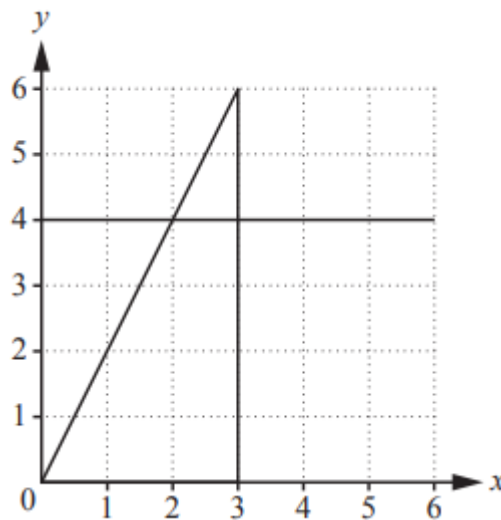
4. W16/qp12/q25



In the diagram, the line $3y + 2x = 13$ meets the axes at A and B.

- Find the coordinates of A. [1]
- The shaded region R is defined by five inequalities. Two of these are $x \leq 6$ and $y \leq 6$. Write down the other three inequalities. [2]
- The point P is in the shaded region R. Given that AP is as large as possible, write down the coordinates of P. [1]

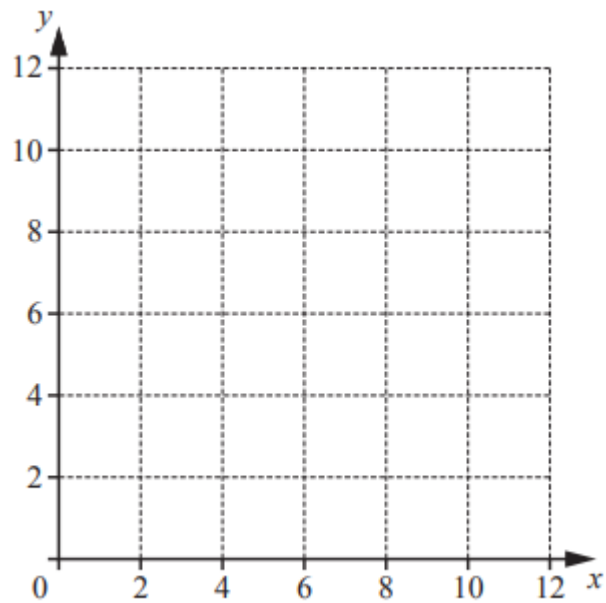
5. S17/qp22/q5(d)



- Draw the graph of $x + 2y = 5$. [2]
- Shade the region defined by these inequalities and label it R.

$$x \leq 3 \quad y \leq 4 \quad y \leq 2x \quad x + 2y \geq 5 \quad [1]$$

6. S18/qp12/q18



The region R is defined by the inequalities

$$2 \leq x \leq 8$$

$$5 \leq y \leq 10$$

$$x + y \geq 10$$

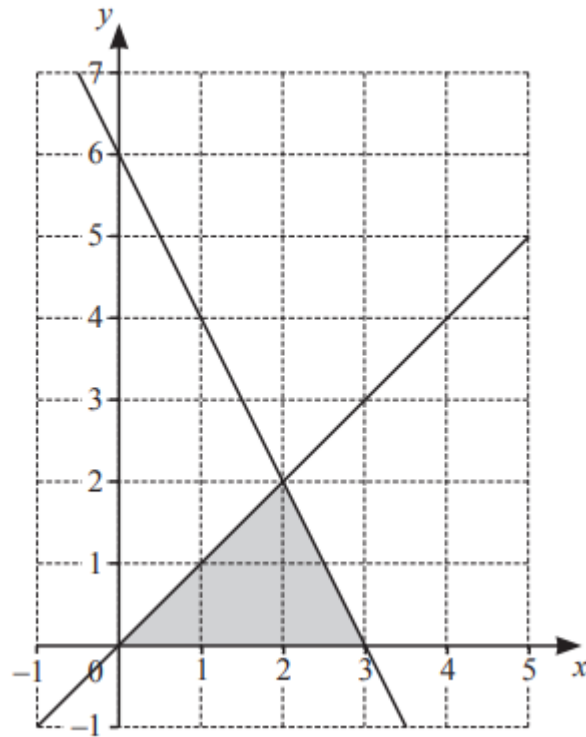
On the diagram, shade and label the region R.

[3]

7. S19/qp21/q6(b)

Find the 3 inequalities which define the region shaded in the diagram.

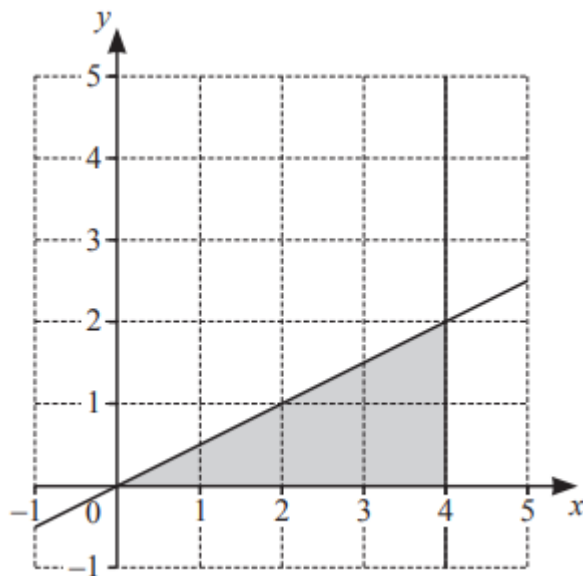
[3]



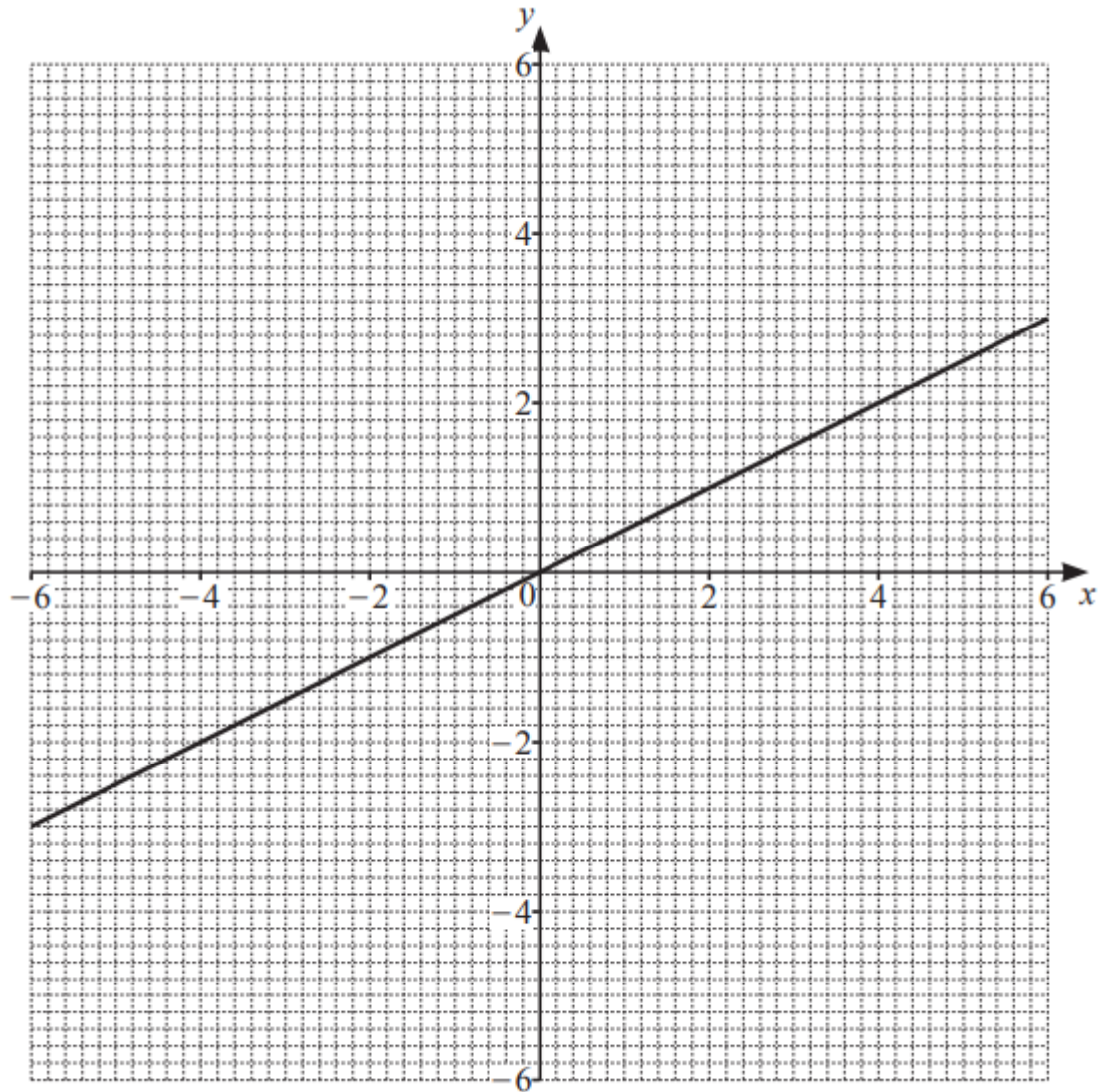
8. S19/qp22/q3(b)

Find the 3 inequalities which define the region shaded in the diagram.

[3]



9. S21/qp11/q20



The line $2y = x$ is drawn on the grid.

- a. On the grid, draw the graph of
 - i. $y = 2$, [1]
 - ii. $y + x = 4$. [1]
- b. On the grid, shade and label the region R, defined by the following inequalities.

$$\begin{aligned} x + y &\leq 4 \\ 2y &\geq x \\ y &\leq 2 \\ x &\geq 0 \end{aligned}$$

[2]

Answer Key:

1.(i) Correct region shaded with 4 correct lines (ii) $-\frac{1}{2}$

2. $y \leq 3, y \geq -x$

3.(a) $x + y \leq 8, 2y \geq x + 4, x \geq 0$ (b) 3

4. $(0, 4\frac{1}{3})$ (b) $x \geq 1, y \geq 2, 3y + 2x \geq 13$ (c) $(6, 2)$

5. Ruled line through $(0, 2.5)$ and $(5, 0)$ (ii) Correct region unambiguously identified

6. Correct region $x = 2, y = 5, x = 10, x + y = 10$.

7. $y \geq 0, x \geq y, 2x + y \leq 6$

8. $y \geq 0, x \leq 4, y \leq \frac{1}{2}x$

9.(a)(i) $y = 2$ (ii) $y + x = 4$ (c) Correct region shaded